Lab 3 Activity 3.1: Configuring a small network (Packet Tracer)

Online: Not Assessed

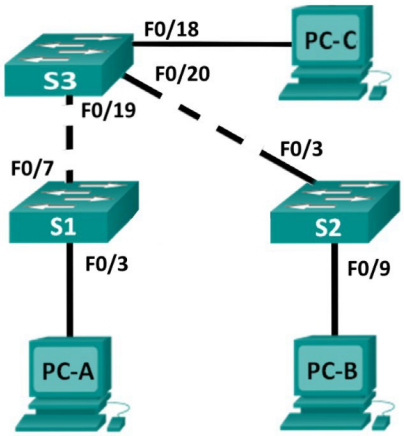
You must use Packet Tracer v8.2.2 to complete this lab.

Please rename your Packet Tracer file to FirstName_LastName_sNumber_Lab3.1.pka

Scoring Scheme

No.	Description	Points
1	PC configuration	15
2	Switch configuration	25-33

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask
PC-A	NIC	192.168.1.33	255.255.255.0
PC-B	NIC	192.168.1.65	255.255.255.0
PC-C	NIC	192.168.1.97	255.255.255.0
S1	VLAN 1	192.168.1.1	255.255.255.0
S2	VLAN 1	192.168.1.2	255.255.255.0
S3	VLAN 1	192.168.1.3	255.255.255.0

Note: please note the '-' in the PC names. Names must be exact

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Objectives

Part 1: Configure Basic Network Devices

Part 2: Verify and Test Network Connectivity

Background / Scenario

Cisco switches have a special interface, known as a Switch Virtual Interface (SVI). The SVI can be configured with an IP address, commonly referred to as the management address. The management address is used for remote access to the switch to display or configure settings.

In this lab, you will build a simple network using Ethernet LAN cabling and access Cisco switches using the console and remote access methods. You will configure basic switch settings, IP addressing, and demonstrate the use of a management IP address for remote switch management. You will also configure each switch to all remote access via ssh the secure terminal program instead of using telnet.

Required Resources

- Packet Tracer 8.2.2

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Part 1: Configure Basic Network Devices

In Part 1, you will set up the network and configure basic settings, such as hostnames, interface IP addresses, and passwords using Packet Tracer. Note that this activity is not assessed but you should upload the completed Packet Tracer file using the workshop upload page to ensure that you are familiar with the uploading of packet tracer files for the upcoming Packet Tracer assessments. The Packet Tracer file for the non-assessed workshop is self-marking and will give feedback in the form of score / completion. The activity is completed correctly when the full score is obtained. You must use the supplied Packet Tracer file for Lab 3 to complete the workshop activity as otherwise you will not be able to use the built in feed-back and scoring system that will also be used for the assessments.

Step 1: Create the network

In Packet Tracer, create the network based on the topology. Refer to Labs 1 and 2 if you need a refresher on how todo this and your journal notes.

Note: For each switch in the topology use the 2960 switch

Step 2: Cable the network.

a. Cable the network as shown in the topology.

b. Establish a console connection to Switch S1 from PC-A.

c. Establish a console connection to Switch S2 from PC-B.

d. Establish a console connection to Switch S3 from PC-C.

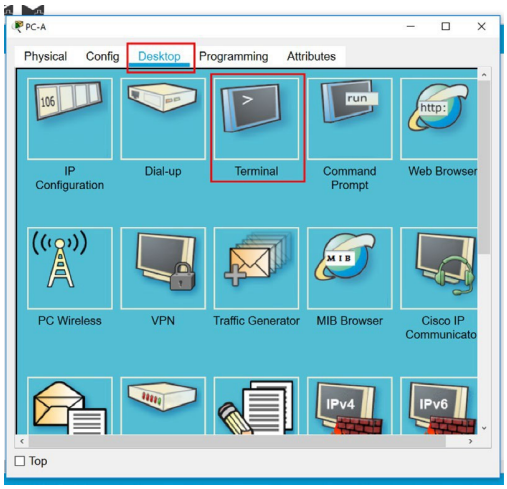
Note: Do not remove the console cables as they form part of the network that will be checked.

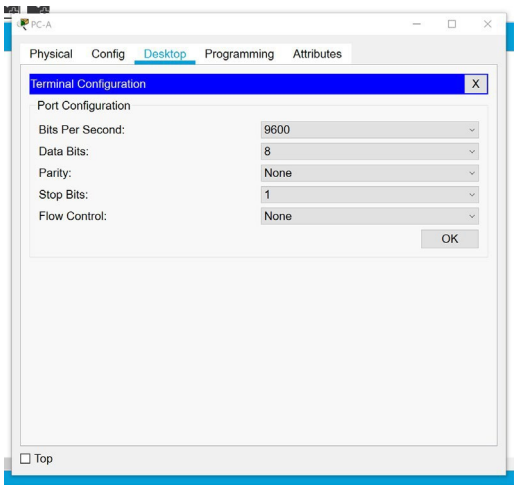
Step 3: Configure basic switch settings.

In this step, you will configure basic switch settings, such as hostname, and configure an IP address for the SVI.

Assigning an IP address on the switch is only the first step. As the network administrator, you must specify how the switch will be managed. Telnet and SSH are two of the most common management methods. However, Telnet is a very insecure protocol. In telnet, all information flowing between the two devices is sent in plaintext. Passwords and other sensitive information can be easily viewed if captured by a packet sniffer. In practice you would not use telnet but would configure SSH instead. In this lab we will use ssh to make connections. In order to configure the switch for use via ssh, we first have to configure it using the console cable.

a. Use the terminal program on PC-A in Packet Tracer to connect to the switch S1. This connection will use the console cable that connects PC-A and S1. On the Terminal configuration window press OK. This will open the Switch Command Line Interface (CLI).





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b. The prompt will be `Switch>`. Enter privileged EXEC mode.

```
Switch> enable
Switch#
```

c. Enter global configuration mode and assign the switch hostname.

```
Switch# configure terminal
Switch(config)# hostname S1
S1(config)#
```

d. Configure the switch password access.

```
S1(config)# enable secret class
S1(config)#
```

e. Prevent unwanted DNS lookups.

```
S1(config)# no ip domain-lookup
S1(config)#
```

f. Configure a login MOTD banner.

Note: Packet Tracer uses strictly text matching when marking the banner. You must write exactly the same banner text including capitalization, spaces and punctuation to be marked correct for this part.

```
S1(config)# banner motd 'Unauthorised access is strictly prohibited.'
```

g. In global configuration mode to set the SVI IP address to allow remote switch management.

```
S1#(config)# interface vlan 1
S1(config-if)# ip address 192.168.1.1 255.255.255.0
S1(config-if)# no shutdown
S1(config-if)# exit
S1(config)#
```

h. Restrict console port access. The default configuration is to allow all console connections with no password needed.

```
S1(config)# line con 0
S1(config-line)# password cisco
S1(config-line)# login
S1(config-line)# logging synchronous
S1(config-line)# exit
S1(config)#
```

Note, do not configure the VTY lines yet. We will configure the VTY lines for the switch to allow remote access in the next section to use ssh.

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i. Encrypt the clear text passwords (ie. cisco). The command for this is:

service password-encryption

Enter the command:

```
S1(config)# service password-encryption
```

Once you have done this, the console and vty passwords will show as encrypted when viewing the running-configuration file:

```
line con 0
 password 7 0822455D0A16
 logging synchronous
 login
```

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Step 4: Setting up SSH

Using Telnet to connect to a network device is a security risk because all information is transmitted in a clear text format. SSH encrypts the session data and provides device authentication, which is why SSH is recommended for remote connections. You will now configure the switch to accept SSH connections over the VTY lines:

The device name and domain are used as part of the cryptographic key when it is generated. Therefore, these names must be entered prior to issuing the crypto key command. The hostname has already been entered (above). We need to supply a domain name:

Step A: Configure the domain for the device.

```
S1(config)# ip domain-name networklab.com
```

Step B: Configure the encryption key method. Please note this command has two parts, first the 'crypto key generate rsa' command, then you will be prompted for the number of bits to use. We will use 1024 bits.

```
S1(config)# crypto key generate rsa

      The name for the keys will be: S1.networklab.com
      Choose the size of the key modulus in the range of 360 to 2048 for
      your General Purpose Keys. Choosing a key modulus greater than 512
      may take a few minutes.

How many bits in the modulus [512]: 1024

      % Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
      (elapsed time was 1 seconds)
      *Jan 28 21:09:29.867: %SSH-5-ENABLED: SSH 1.99 has been enabled

S1(config)#
```

Step C: Configure a local database username. We will use **admin** as username, with **adminpass** as password.

```
S1(config)# username admin privilege 15 secret adminpass
```

Note: A privilege level of 15 gives the user administrator rights.

Step D: Enable SSH on the VTY lines.

a. Enable only SSH on the inbound VTY lines using the **transport input** command.

```
S1(config)# line vty 0 15
S1(config-line)# transport input ssh
```

b. Change the login method to use the local database for user verification and set logging to synchronous.

```
S1(config-line)# login local
S1(config-line)# logging synchronous
S1(config-line)# end
S1#
```

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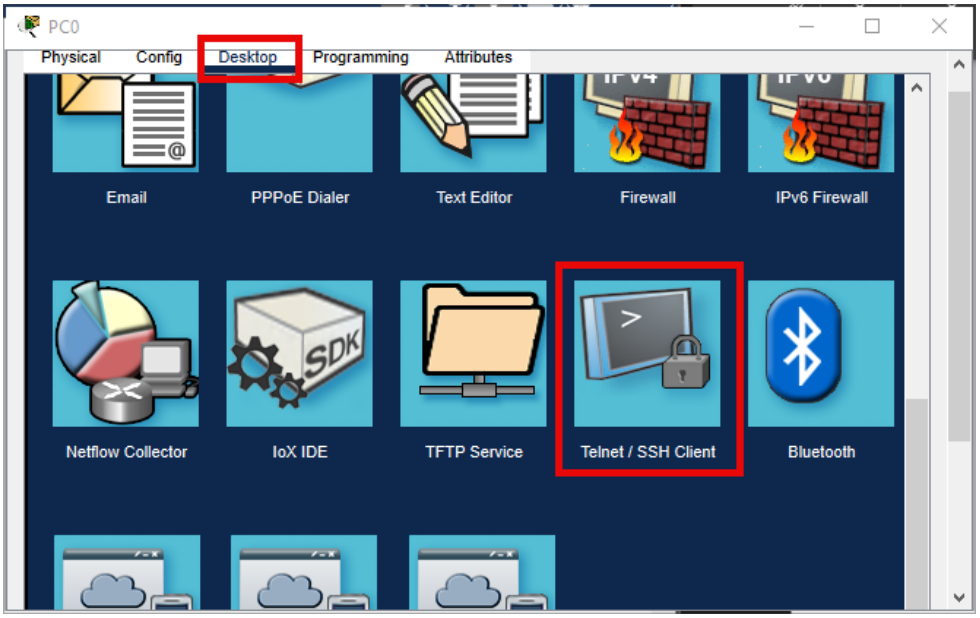
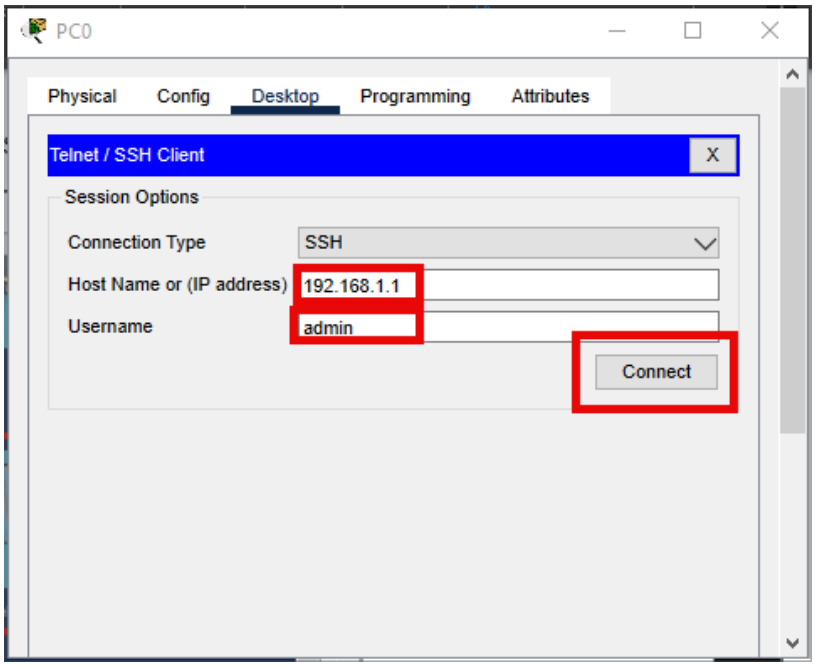
Step E: Testing the SSH connection:

1. Establish an SSH session to S1 using the Telnet / SSH client:



2. Use the IP address for S1, the username **admin:**



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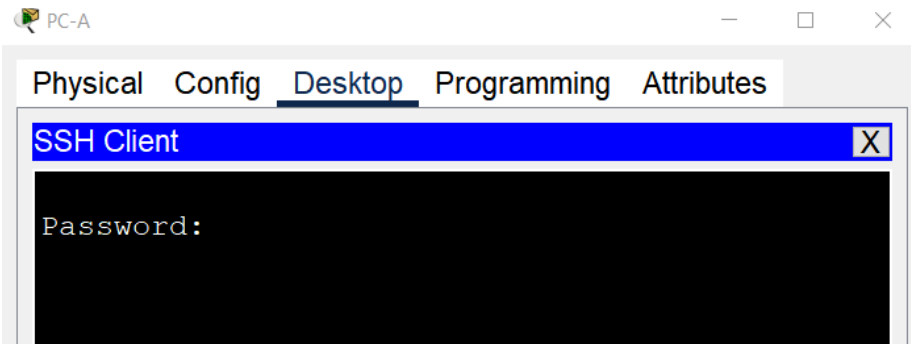
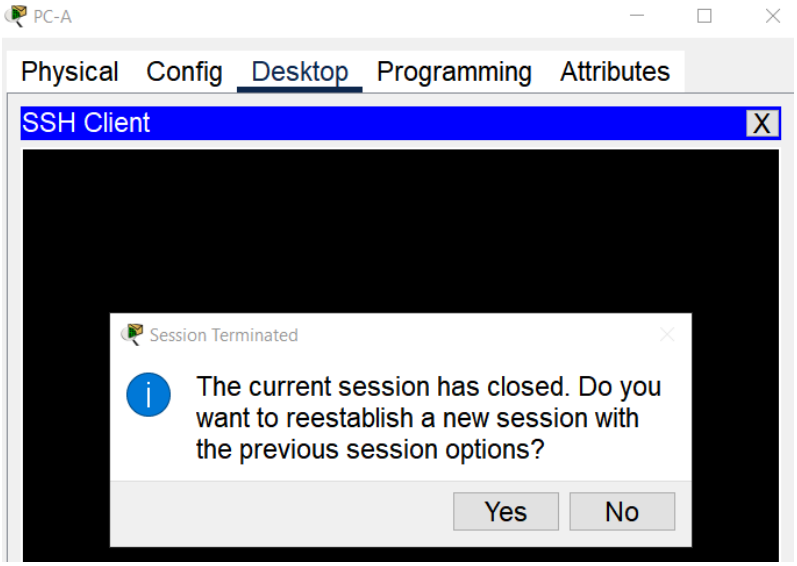
3. On the next screen use the password **adminpass**.



You should now be able to establish an SSH session with S1.

Note: If you try to use the telnet option to connect to S1 you will get an error as telnet is not enable based on the above configuration:



Step F: Repeat the SSH process in steps A to D for the other switches and verify that it has been correctly configured as above in Step E. Note: Only 1 of the VTY lines for each switch will be scored.

Step 5: Configure an IP address on PC-A.

In Packet Tracer, assign the IP address and subnet mask to the PC, as shown in the Addressing Table using the IP configuration program on its desktop.

Step 6: Verify the configurations for PC-A and switch S1

Configure the remaining switches and PCs as per the Addressing Table and the configuration instructions in steps 3 and 4 above.

Display the S1 device configuration.

a. Return to your console connection using the terminal program on PC-A. Issue the **show run** command to display and verify your switch configuration. A sample configuration is shown below. The settings you configured are highlighted in yellow. The other configuration settings are IOS defaults.

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```
S1# show run
Building configuration...

Current configuration : 1508 bytes
!
! Last configuration change at 00:06:11 UTC Mon Mar 1 1993
!
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname S1
!
boot-start-marker
boot-end-marker
!
enable secret 4 06YFDUHH61wAE/kLkDq9BGho1QM5EnRtoyr8cHAUg.2
!
no aaa  new-model
system mtu routing 1500
!
!
no ip domain-lookup
ip domain-name networklab.com
!
username admin secret 5 $1$mERr$89oFbVUY9tU/mdjv3ClG3
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
!
interface FastEthernet0/1
!
interface FastEthernet0/2
<output omitted>
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 ip address 192.168.1.1 255.255.255.0
!
ip http server
ip http secure-server
!
banner motd ^C
```

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```
Unauthorised access is strictly prohibited.^C
!
line con 0
 password 0822455D0A16
 logging synchronous
 login
line vty 0 4
 login local
 transport input ssh
line vty 5 15
 login local
 transport input ssh
!
end
```

b. Verify the status of your SVI management interface. Your VLAN 1 interface should be up/up and have an IP address assigned. Notice that switch port F0/3 is also up because PC-A is connected to it and port 7 is up because it connects to switch S3. Because all switch ports are initially in VLAN 1, by default, you can communicate with the switch using the IP address you configured for VLAN 1.

```
S1# show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
Vlan1                  192.168.1.1     YES manual up                    up
FastEthernet0/1        unassigned      YES unset  down                  down
FastEthernet0/2        unassigned      YES unset  down                  down
FastEthernet0/3        unassigned      YES unset  up                    up
FastEthernet0/4        unassigned      YES unset  down                  down
FastEthernet0/5        unassigned      YES unset  down                  down
FastEthernet0/6        unassigned      YES unset  down                  down
FastEthernet0/7        unassigned      YES unset  up                    up
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  down                  down
FastEthernet0/12       unassigned      YES unset  down                  down
FastEthernet0/13       unassigned      YES unset  down                  down
FastEthernet0/14       unassigned      YES unset  down                  down
FastEthernet0/15       unassigned      YES unset  down                  down
FastEthernet0/16       unassigned      YES unset  down                  down
FastEthernet0/17       unassigned      YES unset  down                  down
FastEthernet0/18       unassigned      YES unset  down                  down
FastEthernet0/19       unassigned      YES unset  down                  down
FastEthernet0/20       unassigned      YES unset  down                  down
FastEthernet0/21       unassigned      YES unset  down                  down
FastEthernet0/22       unassigned      YES unset  down                  down
FastEthernet0/23       unassigned      YES unset  down                  down
FastEthernet0/24       unassigned      YES unset  down                  down
GigabitEthernet0/1     unassigned      YES unset  down                  down
GigabitEthernet0/2     unassigned      YES unset  down                  down
```

Step 7: Repeat Steps 3, 4, 5, and 6 for the remaining PCs and switches

Configure the remaining switches and PCs as per the Addressing Table and the configuration instructions in steps 3, 4, and 5 above.

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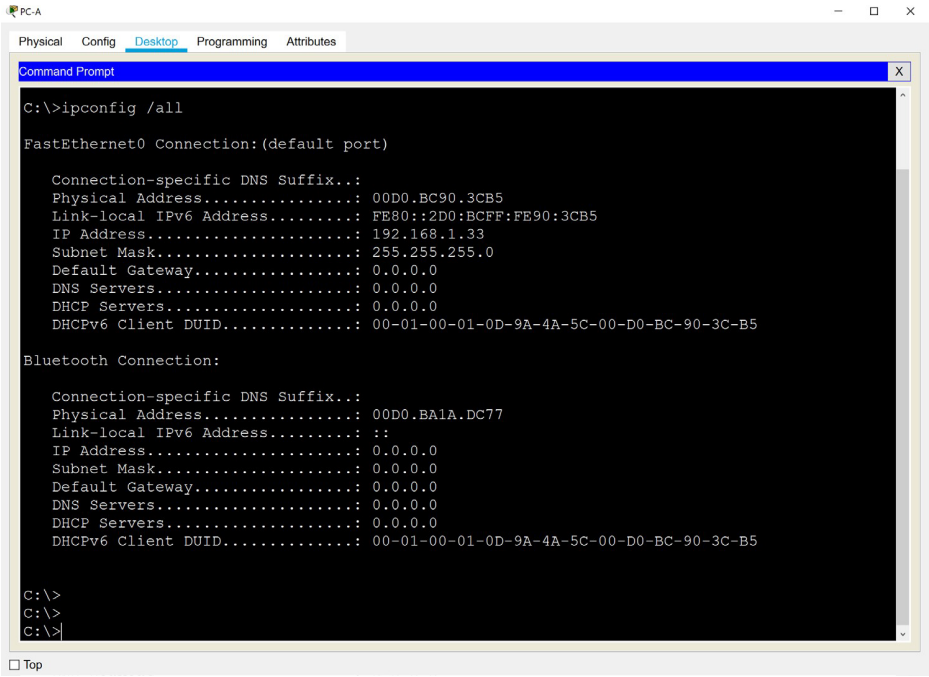
Part 2: Verify and Test Network Connectivity

You will now verify and document the switch configuration, test end-to-end connectivity between the PCs and the switches. Verifying connectivity is an important process and forms part of the troubleshooting activities that you will be performing when configuring a network.

Step 1: Test end-to-end connectivity.

In Packet tracer, open a command prompt window on PC-A. Verify the IP address of PC-A by using the

Ipconfig /all command. This command displays the PC hostname and the IPv4 address information.



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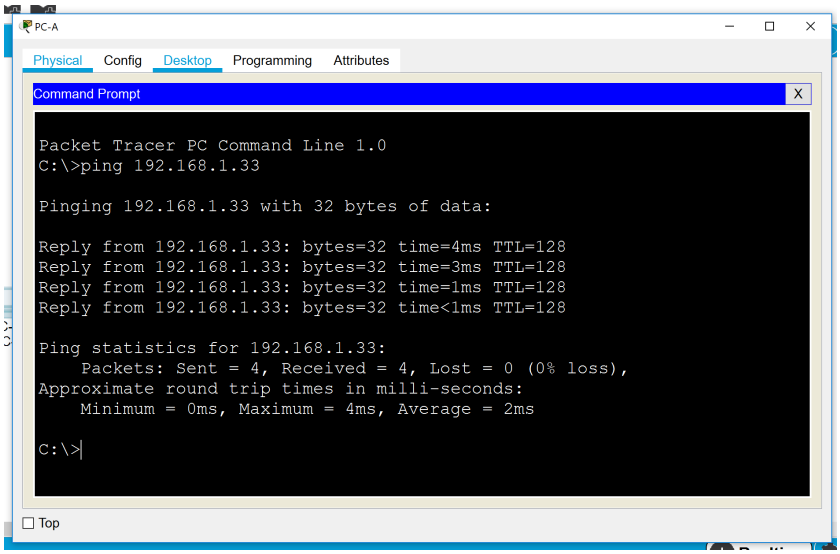
Ping PC-A's address and the management address of S1.

a. Ping the PC-A address first.

```
C:\Users\NetAcad> ping 192.168.1.33
```

Your output should be similar to the following screen:

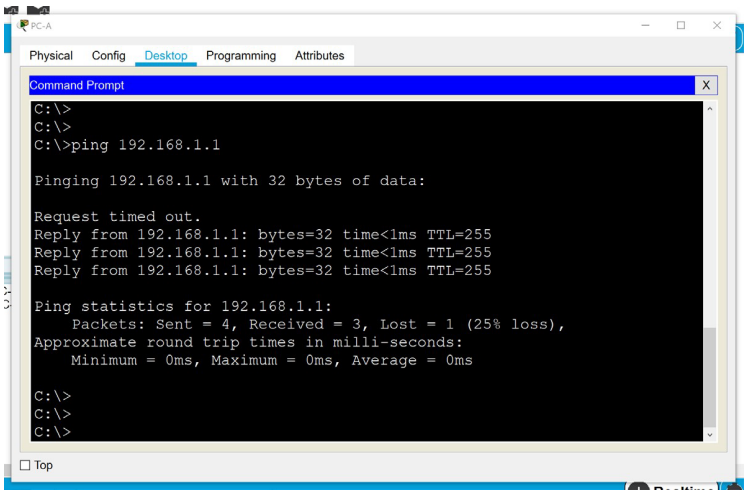


b. From PC-A Ping the SVI management address of S1.

```
C:\Users\NetAcad> ping 192.168.1.1
```

Your output should be similar to the following screen. If no ping results are successful, troubleshoot the basic device configurations. You should check both the physical cabling and IP addressing if necessary.

Note: as in the example below, the first one or two ping results may fail as the connection may still be in the configuration process.



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Step 2: Test and verify the remaining connections.

a. Repeat Steps 1 and 2 to test the connectivity for PC-B to S2, PC-C to S3, PC-D to S4.

b. Test connectivity between PC-A and PC-B

c. Test connectivity between PC-A and PC-C

Reflection

Why must you use a console connection to initially configure the switch? Why not connect to the switch via Telnet or SSH from the start?

_____

Can you also telnet to S1 from PC-B and PC-C?_____

Why?

_____

I. Submit your Packet Tracer file

Submit your Packet Tracer file When you have completed the configuration activities in your Packet Tracer file, please save it as a .pka file (do not save as .pkz), and submit it using the Lab submission point. By submitting it you will be able to become familiar with the submission process used for later Packet Tracer Assessments. We may also use your submissions to give feedback on if your name / student number has been correctly assigned to the file.

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